

Prabhaav Pillai

Computer Science Graduate Student & Software Engineer

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Education

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| MS | Michigan State University , Computer Science <ul style="list-style-type: none">GPA: 4.00/4.00Currently pursuing M.S. Accelerated Computer Science Program (Completed 9 graduate credits during B.S. via MSU Shared Enrollment program) | East Lansing, MI
Jan 2026 – Dec 2026 |
| BS | Michigan State University , Computer Science <ul style="list-style-type: none">GPA: 3.89/4.00 Honors College Minor in MathematicsCompleted 4-year B.S. degree in 2.5 years | East Lansing, MI
Aug 2023 – Dec 2025 |

Experience

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| Extern , AI Document Intelligence Extern (Pfizer Program) <ul style="list-style-type: none">Building an AI document intelligence system to automate data extraction, classification, and compliance risk detection across pharmaceutical records.Developing Python pipelines utilizing Layout-Aware OCR and RAG to query and extract structured insights from complex vendor files.Implementing document classification, dynamic file routing, and open-source LLM workflows integrated with a Gradio chatbot interface. | Remote
July 2026 – present
3 months |
| United Wholesale Mortgage , Software Engineering Capstone Developer (Click for Info: capstone.cse.msu.edu) <ul style="list-style-type: none">Developed full-stack IT Goals Dashboard using React and ASP.NET Core to visualize real-time KPIs for 2,000+ IT professionals.Built EF Core migrations, backend metric aggregation logic, dynamic gauge widgets, and sparkline UI visuals exceeding SonarQube quality standards.Collaborated with product owners to refine yearly/monthly metrics, improving operational transparency across IT teams. | Pontiac, MI
Aug 2025 – Dec 2025
5 months |
| Microsoft TEALS Program , Volunteer CS Teaching Assistant <ul style="list-style-type: none">Instructed 50+ students across two AP Computer Science classrooms in computational thinking, JavaScript, and algorithmic problem-solving.Coordinated with lead teachers to optimize lesson delivery, facilitate hands-on coding labs, and assist high school seniors with college readiness. | Detroit, MI
Aug 2023 – June 2025
1 year 11 months |

Projects

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| Automotive Edge AI System - StarPilot Developer (Click for Info: https://prabhaav.me)
Invited to core dev team for a safety-critical open-source vehicle platform <ul style="list-style-type: none">Appointed to core dev team with repo write access; authored and merged +3.5k lines of C++/Python/GLSL to deploy real-time blind-spot monitoring and a custom GLSL shader Picture-in-Picture (PiP) side camera preview system.Trained custom YOLO26n ONNX models on 27+ hours of crowdsourced driving video using semi-automated CAN-bus radar logging for ground truth.Engineered real-time edge performance on Snapdragon SoCs using zero-copy NV12 buffer slicing and CPU core isolation under 1.2x throttle overhead.Built Flask REST APIs and Galaxy web tools for feature configuration, driving widespread adoption (40K+ views, 185+ upvotes) and maintaining production bug fixes. | July 2026 – present |
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Certifications

- Certified Cloud Practitioner (Score: 981/1000) - Amazon Web Services (July 2026)

Selected Honors

- First Place Award, Education Category - MSU University Undergraduate Research Forum (UURAF, April 2024)
- Researched academic impacts of Large Language Models (LLMs) on student performance using mixed quantitative and qualitative analysis.

Skills

Work Authorization: U.S. Work Authorized via F-1 OPT + STEM OPT (36 months without immediate employer-based sponsorship)

Languages: Python, C++, C#, JavaScript, Java, SQL, GLSL

AI / Machine Learning: PyTorch, YOLOv8/26, Computer Vision, RAG, Layout OCR, ONNX Runtime, Model Optimization

Edge AI & Systems: Embedded AI, Real-time Inference, OpenGL, Linux, CMake, Performance Profiling, Networking

Frameworks & Tools: React, ASP.NET Core, Docker, REST APIs, Git, GitHub, Swagger UI, Fast API, Jupyter Notebook

Cloud & Databases: AWS (Certified Cloud Practitioner), Azure SQL, Entity Framework Core